

Finding speed, distance and time

Three questions from one formula, with the units chosen before the arithmetic.

LESSON 120–122

3 × 40 MIN

MATEMATIKA 6, \$25

S7 12 RATES

LESSON CLOCK

25'

30'

35'

25'

5

Finding the distance	25 min
Finding the speed	30 min
Finding the time	35 min
Hours and minutes	25 min
Homework	5 min

LEARNING OBJECTIVES

- Find any one of s , v , t from the other two.
- Convert times between hours, minutes and seconds.
- Give an answer in hours and minutes where that is natural.
- Check every answer against the sense of the situation.

TEXTBOOK

National	pp. 349–358
Cambridge	Stage 7, pp. 122–126
Workbook	Exercise 12.6

01

Explanation

Finding the distance

$s = vt$

SPEED	TIME	DISTANCE
65 km/h	4 h	260 km
50 km/h	2 h 30 min	125 km
12 km/h	20 min	4 km
15 m/s	40 s	600 m

WRITE THE TIME IN THE UNIT THE SPEED USES

2 h 30 min is 2.5 h; 20 min is $\frac{1}{3}$ h. Doing that conversion first turns every problem into one multiplication.

Finding the speed

$$v = \frac{s}{t}$$

DISTANCE	TIME	SPEED
260 km	4 h	65 km/h
125 km	2.5 h	50 km/h
4 km	20 min	12 km/h
600 m	40 s	15 m/s

A SPEED IN KM/H NEEDS THE TIME IN HOURS

4 km in 20 minutes is 12 km/h, not 0.2 km/h. Dividing by 20 answers a different question.

Finding the time

$$t = \frac{s}{v}$$

DISTANCE	SPEED	TIME	IN HOURS AND MINUTES
260 km	65 km/h	4	4 h
150 km	60 km/h	2.5	2 h 30 min
90 km	80 km/h	1.125	1 h 7.5 min
7 km	5 km/h	1.4	1 h 24 min

TURNING A DECIMAL HOUR INTO MINUTES

Multiply the decimal part by 60: 0.4 h is 24 min, 0.25 h is 15 min. Writing 1.4 h as “one hour forty minutes” is a real and common error.

Hours and minutes

PROBLEM	WORKING	ANSWER
a bus leaves at 8:00 and travels 150 km at 60 km/h	2.5 h	arrives 10:30
a train leaves at 7:20 and takes 1 h 50 min	7:20 + 1:50	arrives 9:10
a journey of 7 km at 5 km/h	1.4 h	1 h 24 min
a walk from 9:15 to 11:00 covering 7 km	1.75 h	4 km/h

ADDING TIMES IS NOT DECIMAL ADDITION

7:20 plus 1 h 50 min is 9:10, because 70 minutes is an hour and ten. Times carry at 60, not at 100.

02

Worked examples

EXAMPLE 1 A cyclist rides at 12 km/h for 20 minutes. How far?

1

$20 \text{ min} = \frac{1}{3} \text{ h.}$

2

$s = 12 \cdot \frac{1}{3}$

3

$= 4 \text{ km.}$

ANSWER

4 km

EXAMPLE 2 A journey of 7 km is walked at 5 km/h. How long does it take?

$$① \quad t = \frac{7}{5} = 1.4 \text{ h.}$$

$$② \quad 0.4 \cdot 60 = 24 \text{ min.}$$

$$③ \quad 1 \text{ h } 24 \text{ min.}$$

Not one hour forty.

ANSWER

1 h 24 min

EXAMPLE 3 A bus leaves at 8:00 and covers 150 km at 60 km/h. When does it arrive?

$$① \quad t = \frac{150}{60} = 2.5 \text{ h.}$$

$$② \quad 2.5 \text{ h is } 2 \text{ h } 30 \text{ min.}$$

$$③ \quad 8:00 + 2:30 = 10:30.$$

ANSWER

10:30

03 Interactive model

Use a real bus timetable: pupils compute the average speed between two stops and compare it with the road speed limit, which is a genuinely interesting comparison.

Which one is wanted?

65 km/h for 4 h:

16.25 km

260 km

69 km

260 km/h

50 km/h for 2 h 30 min:

100

125

150

250

12 km/h for 20 min:

240

4

0.6

36

4 km in 20 min:

0.2 km/h

12 km/h

80 km/h

4 km/h

150 km at 60 km/h takes:

2.5 h

90 h

2.3 h

9000 h

1.4 h in hours and minutes:

1 h 40 min

1 h 24 min

1 h 4 min

14 min

7:20 plus 1 h 50 min:

8:70

9:10

8:10

9:70

7 km from 9:15 to 11:00:

4 km/h

7 km/h

1.75 km/h

12.25 km/h

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Formula	Formula	Формула
To rearrange	O'zgartirib yozish	Выразить
Hours and minutes	Soat va daqiqa	Часы и минуты
Decimal hours	O'nli soat	Часы в десятичной форме
Arrival time	Yetib borish vaqti	Время прибытия
Departure	Jo'nash	Отправление
Journey	Sayohat	Поездка
Sense check	Mantiqiy tekshirish	Проверка по смыслу

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To find the distance you:

divide

multiply

subtract

add

To find the time you:

$s \cdot v$

$s \div v$

$v \div s$

$s + v$

20 minutes in hours is:

0.2

$\frac{1}{3}$

0.5

2

0.25 h in minutes is:

90 km at 80 km/h takes:

A bus leaving at 8:00 and taking 2 h 30 min arrives at:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 65 km/h for 4 h

ANSWER 260 km

2. 50 km/h for 2.5 h

ANSWER 125 km

3. 15 m/s for 40 s

ANSWER 600 m

4. 260 km in 4 h

ANSWER 65 km/h

5. 125 km in 2.5 h

ANSWER 50 km/h

6. 260 km at 65 km/h

ANSWER 4 h

7. 150 km at 60 km/h

ANSWER 2.5 h

Medium · the standard question

7 PROBLEMS

1. 12 km/h for 20 min

ANSWER 4 km

2. 4 km in 20 min

ANSWER 12 km/h

3. 7 km at 5 km/h

ANSWER 1 h 24 min

4. 90 km at 80 km/h

ANSWER 1 h 7.5 min

5. A bus leaves 8:00, 150 km at 60 km/h: arrival

ANSWER 10:30

6. 7:20 plus 1 h 50 min

ANSWER 9:10

7. 7 km from 9:15 to 11:00

ANSWER 4 km/h

Hard · stretch and reason

7 PROBLEMS

1. A train leaves 6:45 and takes 3 h 40 min: arrival

ANSWER 10:25

2. A journey of 340 km at 85 km/h

ANSWER 4 h

3. A journey of 200 km taking 2 h 40 min

ANSWER 75 km/h

4. How far in 1 h 45 min at 60 km/h?

ANSWER 105 km

5. A car covering 45 km in 30 min

ANSWER 90 km/h

6. A walk of 11 km at 4 km/h

ANSWER 2 h 45 min

7. Why is 1.4 h not 1 h 40 min?

ANSWER The decimal part is 0.4 of an hour, which is 24 minutes

07 Homework

Homework — 5 tasks

Convert to hours before dividing, and back to minutes at the end.

1. A car travels 3 h 15 min at 80 km/h. Find the distance.
2. A journey of 96 km takes 1 h 20 min. Find the speed.
3. How long does 220 km take at 55 km/h?
4. A bus leaves at 7:40 and takes 2 h 35 min. When does it arrive?
5. A walker covers 9 km at 4 km/h. Give the time in hours and minutes.